

Uber Fare Fetch Flow - Architecture & Component Responsibilities

Overview

This document describes a Flutter application flow that authenticates a user with Uber in a WebView, reuses the authenticated session in a background WebView, loads the Product Selection page, extracts fare information using injected JavaScript, and displays the results in Flutter.

Components

Flutter App: UI, navigation, WebView control, cookie persistence, JSON parsing.

WebView: Hosts Uber login and authenticated pages.

Uber Auth: <https://auth.uber.com/v2/> authenticates the user.

Product Selection: <https://www.uber.com/go/product-selection?>... loads ride options.

GraphQL API: <https://m.uber.com/go/graphql> returns ride/product data for the authenticated session.

Location API: <https://www.uber.com/api/pudoLocationSearch?localeCode=en> resolves pickup/drop locations.

Injected JavaScript: Extracts display name and price from the rendered page.

JavaScript Channel: Transfers extracted JSON to Flutter.

Detailed Request / Response Flow

1. Flutter checks for stored session cookies.
2. If unavailable, open <https://auth.uber.com/v2/> in a WebView.
3. User completes authentication.
4. WebView stores Uber session cookies; Flutter persists them.
5. User enters pickup and drop.
6. Location information is resolved (e.g. via the location search endpoint) into addresses, coordinates, and place identifiers.
7. Flutter builds the Product Selection URL and loads it in a hidden WebView after restoring cookies.
8. The Product Selection page issues authenticated background requests including <https://m.uber.com/go/graphql>. The browser automatically attaches the stored cookies.
9. Uber returns ride/product information which is rendered on the page.
10. Injected JavaScript extracts ride display names and prices.
11. JavaScript posts JSON to Flutter through a JavaScriptChannel.
12. Flutter parses the data and navigates to the Fare Result screen.

Responsibilities

Flutter: session management, navigation, rendering.

WebView: authenticated browser context.

Uber Services: authentication, location resolution, fare computation.

Injected JS: DOM extraction only.

Fare Result Screen: display ride name and price.